

Omkar Thawakar

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Abu Dhabi, UAE

RESEARCH INTEREST

Passionate researcher in video retrieval, multimodal learning, and large language models (LLMs), with hands-on experience in video instance segmentation, dense retrieval, and large-scale multimodal datasets. Interested in contributing innovative ideas, technical knowledge, and dedication to organizations that foster collaborative research.

EXPERIENCE

- **Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)**  May 2021 - Jul 2023
Researcher
Abu Dhabi, UAE
 - Worked on Composed Video Retrieval, where we proposed the idea of using dense video description to retrieve the target video with desired compositional changes.
 - Worked on video instance segmentation with transformers. (SOTA performance, ECCV 2022)
 - Implemented 3D Mitochondria Instance Segmentation with Spatio Temporal Transformers with more than 3% overall improvement in segmentation accuracy over recent best. (SOTA performance, MICCAI-2023)
 - Worked on open-world video instance segmentation where we presented the finding on using shallow features for novel object detection. (IJCV-2023)
- **Chefling India Pvt Ltd**  Feb 2020 - May 2021
Machine Learning Engineer
Bangalore, India
 - Worked on creating and deploying own recipe scraping service with Machine Learning model supported by python scrapy tools, increasing the accuracy of system to 87% for complex websites.
 - Implemented recipe tagging service for raw recipe data comprising the 70+ variety of tags, reducing the dependency on external services and saving the expenditure cost by 92%.
- **Indian Institute of Technology Ropar (IIT-Ropar)**  Jan 2019 - Feb 2021
Research Assistant
Ropar, India
 - Worked on vision problems such as image and video super resolution, moving object segmentation, and image restoration (hazy, foggy, lighting clutter ... etc).
 - Developed innovative AI solutions for educational and commercial applications, enhancing learning experiences.

EDUCATION

- **Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)** Aug 2023 - May 2026
PhD, Computer Vision
Abu Dhabi, UAE
 - Thesis: Video Understanding with LMM's, Advisor : [Prof. Fahad Khan](#)
 - Research Areas: Multi-Modal Learning, Vision-Language Modeling, Efficient Large Language Models, MultiLingual LLM, LLM Agents, Video Understanding, Image/Video Detection & Segmentation, etc.
 - GPA: 3.74/4.00
- **Shri Guru Gobind Singhji Institute of Engineering and Technology (SGGSIE&T)** Jun 2015 - Jul 2019
Undergrad, Computer Science
Nanded, India
 - Courseworks Taken: Neural Networks, Machine Learning, Data Structure, Algorithms, Computer Networking, Operating Systems, Cryptography, Distrubuted Systems, Advanced Computer Programming, etc.
 - GPA: 7.7/10.0

PROJECTS

- **Lawa.AI: [Empower your business websites with the power of LLM's]** May 2024 - Present
Tools: [LLM, Agentic Ecosystem, Knowledge Graph, MultiModal & Document Retrieval] 
 - Lawa.AI helps businesses optimize their website & internal organization by providing data-driven insights.
 - By operating directly on the device, lawa.ai ensures data privacy and security for sensitive information.
 - Lawa.ai continually improves through analyzing user interactions to provide more relevant answers over time.
 - Lawa.ai supports multilingual conversations, enabling businesses to reach a global audience.
- **Nutrigenics.Care: [Empowering Health and Wellness for Everyone, Everywhere.]** Jan 2024 - Present
Tools: [LLM, Recommendation System, MongoDB, Django, Flask] 
 - Utilizes advanced AI to instantly track and analyze the nutritional content of meals, ensuring healthier choices.
 - Enables dietitians to track patient meal intake through user-friendly, real-time charts.
 - Engages patients with a user-friendly interface that simplifies the logging process and encourages regular interaction, making it easy to track health data and stay involved in their care.
 - Utilizes advanced AI to provide intelligent recipe suggestions tailored to dietary needs and preferences.

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [S.1] Omkar Thawakar, Shravan Venkatraman, Ritesh Thawkar, *et al.* (2025). **EvoLMM: Self-Evolving Large Multimodal Models with Continuous Rewards**. *Under Review*
- [S.2] Omkar Thawakar, Dmitry Demidov, *et al.* (2025). **CoVR-R: Reason-Aware Composed Video Retrieval**. *Under Review*
- [S.3] A Shaker, ..., Omkar Thawakar, *et al.* (2025). **Mobile-O: Unified Multimodal Understanding and Generation on Mobile Device**. *Under Review*
- [C.1] Omkar Thawakar, Dmitry Demidov, Ritesh Thawkar, *et al.* (2025). **Beyond Simple Edits: Composed Video Retrieval with Dense Modifications**. In ICCV-2025
- [C.2] Omkar Thawakar, *et al.* (2025). **Llamav-o1: Rethinking step-by-step visual reasoning in llms**. In ACL-2025 Huggingface Downloads: 50K+
- [C.3] Sara Ghaboura,... Omkar Thawakar, *et al.* (2025). **Time Travel: A Comprehensive Benchmark to Evaluate LMMs on Historical and Cultural Artifacts**. In ACL-2025
- [C.4] Omkar Thawakar, A Vayani *et al.* (2024). **Mobillama: Towards accurate and lightweight fully transparent GPT**. In ICLR-2025, SLLM Spotlight (Top-2%), Huggingface Downloads: 150K+.
- [C.5] A Vayani,... Omkar Thawakar, *et al.* (2024). **All Languages Matter: Evaluating LMMs on Culturally Diverse 100 Languages**. In CVPR-2025 (Highlight)
- [C.6] Omkar Thawakar, *et al.* (2024). **Composed Video Retrieval via Enriched Context and Discriminative Embeddings**. In CVPR-2024
- [C.7] Sara Ghaboura, Ahmed Heakl, Omkar Thawakar, *et al.* (2024). **CAMEL-Bench: A Comprehensive Arabic LMM Benchmark**. In NAACL-2025
- [C.8] Wafa Alghallabi, .. Omkar Thawakar, *et al.* (2024). **Fann or Flop: A Multigenre, Multiera Benchmark for Arabic Poetry Understanding in LLMs**. In EMNLP-2025 (Main-Track)
- [S.4] Komal Kumar, Tajamul Ashraf, Omkar Thawakar, *et al.* (2024). **LLM Post-Training: A Deep Dive into Reasoning Large Language Models**. In *Arxiv-2025*
- [J.1] Omkar Thawakar, *et al.* (2024). **Video instance segmentation in an open-world**. *International Journal of Computer Vision (IJCV) 2024*, DOI:s11263-024-02195-4
- [C.8] Omkar Thawakar, *et al.* (2024). **XrayGPT: Chest radiographs summarization using large medical vision-language models**. In ACL-W-2024
- [C.9] Omkar Thawakar, *et al.* (2023). **3D Mitochondria Instance Segmentation with Spatio-Temporal Transformers**. In MICCAI-2023
- [C.10] SS Mullappilly, A Shaker, Omkar Thawakar, *et al.* (2023). **Arabic Mini-ClimateGPT: A Climate Change and Sustainability Tailored Arabic LLM**. In ACL-W-2023
- [C.11] Omkar Thawakar, *et al.* (2023). **Fast Video Instance Segmentation via Recurrent Encoder-Based Transformers**. In CAIP-2023
- [C.12] Omkar Thawakar, *et al.* (2024). **Video instance segmentation via multi-scale spatio-temporal split attention transformer**. In ECCV-2022
- [P.1] Omkar Thawakar, *et al.* (2024). **System and method for video instance segmentation via multi-scale spatio-temporal split attention transformer**. US Patent App. 17/983,841.

ENTREPRENEURSHIP EXPERIENCE

- **Cofounder and Acting CEO Lawa.AI** July 2024 - Present
Lawa.AI : Empower your business with the power of Agentic-AI 
 - Led end-to-end R&D for cutting-edge AI products, driving innovation across multimodal LLMs, retrieval pipelines, and production-grade Agentic-AI deployments ensuring reliability, accuracy, and measurable impact.
 - Achieved early traction with a \$70K seed grant, growing the venture to \$130K+ yearly revenue.
- **Cofounder and CTO Nutrigenics.Care** Jan 2024 - Present
Nutrigenics.Care : Empowering Health and Wellness for Everyone, Everywhere. 
 - Led end-to-end development of the Nutrigenics platform, including R&D for the Nutrition-GPT AI engine.
 - Collaborated with two healthcare partners to enhance model accuracy, clinical relevance, and platform usability.
 - Raised \$100K in grant funding through competitive pitch events.

SKILLS

- **Programming Languages:** Python, R, Java, C
- **Web Technologies:** Flask, Django, React-JS, MD-Bootstrap, PHP
- **Database Systems:** MongoDB, SQLite, BigQuery
- **Data Science & Machine Learning:** PySpark, Pytorch, Tensorflow, NumPy, Pandas, Matplotlib, Scikit-learn
- **Cloud Technologies:** AWS, Azure, Google Cloud Platform

HONORS AND AWARDS

- **Spotlight: ICLR-W2025.** Feb 2025
MobiLlama selected as spotlight research paper (Top 2%) at ICLR-W2025 SLLM (Sparsity in Large Language Models). 
- **Winner: Khalifa Fund Entrepreneurship Competition** Dec 2024
Secured first place in the Entrepreneurship Competition with 250K AED startup grant support held at Abu Dhabi business week. 
- **Winner: Sandook-Al-Watan Student Pitch competition** Nov 2024
Secured first place in student pitch competition with 30K AED startup grant support. 
- **Lawa.AI featured in Panel Discussion at Innovation Xchange in ITU-WTSA 2024** Nov 2024
Represented Lawa.AI as emerging startup from UAE at Innovation Xchange program during WTSA event at New Delhi. 
- **Microsoft Founders Hub Grant** July 2024
Nutrigenics.Care selected for Microsoft Founders Hub Program with \$150K of Technical Support. 
- **Winner: Pitch Day with Microsoft** May 2024
Nutrigenics.Care secured first place in Pitch Day with Microsoft organized by Incubation & Entrepreneurship Center MBZUAI. 
- **Winner: Best Project Award for 3D Computer Vision Coursework.** Nov 2023
Worked on 3D Reconstruction using NeRF without using camera parameters for 3D Computer Vision course by Dr. Hao Li. 
- **Winner: Best Project Award for B.Tech Thesis.** May 2019
Worked on Video Super-Resolution using Recurrent Generative Adversarial Network supervised by Dr. U.V. Kulkarni. 
- **Winner: Best Reasarch Paper award in International Conference (IC3NS-2018).** May 2018
Worked on the Application of Machine Learning Algorithms for IPM device. 
- **Winner: Maze Solver Robot Competition for 3 consecutive years 2016-2018 at Pragya SGGS (Tech Event).** Feb 2018
Developed the Maze Solver Robot using Reinforcement Learning that learns the path by itself using Q-Table. 
- **Winner: Science Exhibition Competition during High School.** Aug 2012
Developed the earthquake detector alarm system using Potentiometer & Relay. 

VOLUNTEER EXPERIENCE

- **Teaching Assistant** Jan 2024 - Dec 2024
MBZUAI, Department of Computer Vision 
 - Assisted professors in preparing and delivering lectures, grading assignments, and maintaining student records.
 - Provided one-on-one and group tutoring sessions for the coursework project and helped the student streamline the project deliverables.
- **Reviewer for Conferences such as CVPR, ICCV, ECCV etc** July 2022 - Present 
IEEE, CVF, Springer
 - Provided detailed, constructive feedback to authors to improve the quality and clarity of their submissions.
 - The significance, originality, and technical soundness of the proposed methodologies and experimental results were assessed. 25 plus research papers were reviewed from top-tier venues.

REFERENCES

1. **Prof. Fahad Khan**
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